**User Guide: Automated MISRA Static Analysis Pipeline**

Create a dedicated folder called "static_anlysis" in your repository **root** to hold the python script and template dax file.

<u>The following is ONLY for dax file execution after generation by python script : -</u>

- Enable this macro in CMake File (Change OFF to ON for execution after generation)(Line 6 in included CMakeLists
  file)

```
option(ENABLE_MISRA_EXECUTION "Run AbsInt a3 after generating DAX files" OFF)
```

- Open the python script and locate Line 21. Ensure the path perfectly matches where AbsInt is installed on your computer.

```
A3_EXE = "a3"
```

- Line 83 is the command the python script uses to execute the dax file using command prompt. (change if needed)

```
cmd = f'{safe_exe} --batch --log-file "{log_file}" "{output_dax}"' #This line is the command
```

The included CMakeLists file includes a function to invoke the python script from lines 1 to 23. Include this at the start your existing CMakeLists file.

```
# CMAKE PYTHON BRIDGE FUNCTION
function(add_misra_target MODULE_NAME SOURCES_VAR INCLUDES_VAR OUTPUT_DIR)
```

After this **create a source, include, destination variable** as done below **for each module** and call the function "add_misra_target" at the end of CMakeLists file.

```
#Module 1 :-
set(MAIN
    "${CMAKE_CURRENT_SOURCE_DIR}/source/src/main.c" #Source File Paths
)
set(MAIN_INC
    "${CMAKE_CURRENT_SOURCE_DIR}/source/include"  #Include File Paths
)

set(MAIN_DST "${CMAKE_CURRENT_SOURCE_DIR}/SA_Reports/main") #Destination Path for .dax file and reports folder

add_misra_target("MAIN" MAIN MAIN_INC "${MAIN_DST}") #Function call to python
```

Every time the project is built the CMakeLists file will generate dax file and reports for the mentioned modules in the users desired locaiton. (Reports are only generated when the execution of dax file after

generation is truned on using the macro in CMakeLists file.)